Signals and Systems
Assignment-4: Solution

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Exercise 1

Find the total area under the function: $x(t) = 10\text{sinc}\left(\frac{t-3}{5}\right)$.

- *Step 1:* Calculate the Fourier transform of $x(t)$:

$$\begin{aligned}
\text{sinc}(t) &\rightleftharpoons \text{rect}(f) \\
10\text{sinc}(t) &\rightleftharpoons 10\text{rect}(f) \\
t \to \frac{t}{5} \Rightarrow 10\text{sinc}\left(\frac{t}{5}\right) &\rightleftharpoons 50\text{rect}(5f) \\
t \to t-3 \Rightarrow 10\text{sinc}\left(\frac{t-3}{5}\right) &\rightleftharpoons 50\text{rect}(5f)e^{-j2\pi 3f} \\
&= 50\text{rect}(5f)e^{-j6\pi f}
\end{aligned}$$

- *Step 2:*

$$\int_{-\infty}^{+\infty} x(t)\mathrm{d}t = X(0) = 50\text{rect}(0)e^{0} = 50$$

Exercise 2 (1)

Find the Fourier series coefficients of
$x(t) = \mathrm{rect}\,(4t) * \mathrm{comb}\,(2t)$.

- *Step 1:* Calculate the Fourier transform of $\mathrm{rect}\,(4t)$.

$$\mathrm{rect}(t) \rightleftharpoons \mathrm{sinc}(f)$$

$$\mathrm{rect}(4t) \rightleftharpoons \frac{1}{4}\mathrm{sinc}\left(\frac{f}{4}\right)$$

- *Step 2* Calculate the Fourier transform of $\mathrm{comb}\,(2t)$.

$$\mathrm{comb}(t) \rightleftharpoons \mathrm{comb}(f)$$

$$\mathrm{comb}(2t) \rightleftharpoons \frac{1}{2}\mathrm{comb}\left(\frac{f}{2}\right)$$

Given that $\mathrm{comb}(af) = \frac{1}{|a|}\sum_{k\in\mathbb{Z}} \delta(f - \frac{k}{a})$:

$$\mathrm{comb}(2t) \rightleftharpoons \frac{1}{2}\left(2\sum_{k\in\mathbb{Z}} \delta(f-2k)\right) = \sum_{k\in\mathbb{Z}} \delta(f-2k)$$

Exercise 2 (2)

- *Step 3:*

$$
\begin{aligned}
x(t) &= \mathrm{rect}\,(4t) * \mathrm{comb}\,(2t) \\
\Rightarrow X(f) &= \mathcal{F}\left[\mathrm{rect}\,(4t)\right]\mathcal{F}\left[\mathrm{comb}\,(2t)\right] \\
&= \frac{1}{4}\mathrm{sinc}\left(\frac{f}{4}\right)\sum_{k\in\mathbb{Z}}\delta(f-2k) \\
&= \frac{1}{4}\sum_{k\in\mathbb{Z}}\mathrm{sinc}\left(\frac{f}{4}\right)\delta(f-2k) \\
&= \frac{1}{4}\sum_{k\in\mathbb{Z}}\mathrm{sinc}\left(\frac{2k}{4}\right)\delta(f-2k) \quad = \frac{1}{4}\sum_{k\in\mathbb{Z}}\mathrm{sinc}\left(\frac{k}{2}\right)\delta(f-2k)
\end{aligned}
$$

Given that for periodic signals $X(f) = \sum_{k\in\mathbb{Z}} X[k]\delta(f - kf_0)$:

$f_0 = 2$ (fundamental frequency of the periodic signal $x(t)$)

$$X[k] = \frac{1}{4}\mathrm{sinc}\left(\frac{k}{2}\right) \quad k \in \mathbb{Z}$$

Exercise 2 (3)

Second Method:

- $x(t)$ can be written as:

$$x(t) = \frac{1}{2}\mathrm{rect}(4t) * \frac{1}{1/2}\mathrm{comb}\left(\frac{t}{1/2}\right)$$

  which corresponds to the periodic repetition of the rectangular pulse $\frac{1}{2}\mathrm{rect}(4t)$ of duration $w = \frac{1}{8} - \frac{-1}{8} = \frac{1}{4}$ with a period $T = \frac{1}{2}$.

- Consequently (from the lecture on the FS):

$$\begin{aligned} X[k] &= \frac{1}{2}\frac{w}{T}\mathrm{sinc}\left(\frac{w}{T}k\right) \\ &= \frac{1}{2}\left[\frac{1}{2}\mathrm{sinc}\left(\frac{1}{2}k\right)\right] = \frac{1}{4}\mathrm{sinc}\left(\frac{k}{2}\right) \quad k \in \mathbb{Z} \end{aligned}$$

Exercise 3 (1)

Find the Fourier series coefficients of
$x(t) = \text{sinc}\left(\frac{t}{2}\right) * \frac{1}{10}\text{comb}\left(\frac{t}{10}\right)$.

- *Step 1:* Calculate the Fourier transform of $\text{sinc}\left(\frac{t}{2}\right)$.

$$\text{sinc}(t) \rightleftharpoons \text{rect}(f)$$
$$\text{sinc}\left(\frac{t}{2}\right) \rightleftharpoons 2\text{rect}(2f)$$

- *Step 2* Calculate the Fourier transform of $\frac{1}{10}\text{comb}\left(\frac{t}{10}\right)$.

$$\text{comb}(t) \rightleftharpoons \text{comb}(f)$$
$$\frac{1}{10}\text{comb}(t) \rightleftharpoons \frac{1}{10}\text{comb}(f)$$
$$\frac{1}{10}\text{comb}\left(\frac{t}{10}\right) \rightleftharpoons \frac{1}{10}10\text{comb}(10f) = \text{comb}(10f)$$
$$= \frac{1}{10}\sum_{k\in\mathbb{Z}} \delta\left(f - \frac{k}{10}\right)$$

Exercise 3 (2)

- *Step 3:*

$$
\begin{aligned}
x(t) &= \operatorname{sinc}\left(\frac{t}{2}\right) * \frac{1}{10}\operatorname{comb}\left(\frac{t}{10}\right) \\
\Rightarrow X(f) &= \mathcal{F}\left[\operatorname{sinc}\left(\frac{t}{2}\right)\right]\mathcal{F}\left[\frac{1}{10}\operatorname{comb}\left(\frac{t}{10}\right)\right] \\
&= \frac{1}{5}\operatorname{rect}(2f)\sum_{k\in\mathbb{Z}}\delta\left(f-\frac{k}{10}\right) \\
&= \frac{1}{5}\sum_{k\in\mathbb{Z}}\operatorname{rect}(2f)\,\delta\left(f-\frac{k}{10}\right) \\
&= \frac{1}{5}\sum_{k\in\mathbb{Z}}\operatorname{rect}\left(2\frac{k}{10}\right)\delta\left(f-\frac{k}{10}\right) \\
&= \frac{1}{5}\sum_{k\in\mathbb{Z}}\operatorname{rect}\left(\frac{k}{5}\right)\delta\left(f-\frac{k}{10}\right)
\end{aligned}
$$

Exercise 3 (3)

- Given that for periodic signals $X(f) = \sum_{k\in\mathbb{Z}} X[k]\delta(f - kf_0)$:

$$f_0 = \frac{1}{10} \quad \text{(fundamental frequency of the periodic signal } x(t))$$

$$X[k] = \frac{1}{5}\mathrm{rect}\left(\frac{k}{5}\right) \quad k \in \mathbb{Z}$$

- Given that $\mathrm{rect}(x) = 1$ for $|x| < \frac{1}{2}$ (and 0 elsewhere), then:

$$X[k] = \begin{cases} \frac{1}{5}, & \left|\frac{k}{5}\right| < \frac{1}{2} \Rightarrow |k| < \frac{5}{2} \Rightarrow k = 0, \pm 1, \pm 2; \\ 0, & \text{otherwise.} \end{cases}$$

Exercise 4 (1)

Determine the Fourier transform of $\sin(2t-1)$.

$$\sin(t) = \sin\left(2\pi\left(\frac{1}{2\pi}\right)t\right) \Rightarrow f_0 = \frac{1}{2\pi}$$

$$\sin(t) \rightleftharpoons \frac{j}{2}\delta\left(f+\frac{1}{2\pi}\right) - \frac{j}{2}\delta\left(f-\frac{1}{2\pi}\right)$$

$$t \to t-1 \to \sin(t-1) \rightleftharpoons \left[\frac{j}{2}\delta\left(f+\frac{1}{2\pi}\right) - \frac{j}{2}\delta\left(f-\frac{1}{2\pi}\right)\right] e^{-j2\pi f}$$

$$t \to 2t \to \sin(2t-1) \rightleftharpoons \frac{1}{2}\left[\frac{j}{2}\delta\left(\frac{f}{2}+\frac{1}{2\pi}\right) - \frac{j}{2}\delta\left(\frac{f}{2}-\frac{1}{2\pi}\right)\right] e^{-j2\pi\frac{f}{2}}$$

Given that $\delta(a(t-t_0)) = \frac{1}{|a|}\delta(t-t_0)$:

$$\sin(2t-1) \rightleftharpoons \frac{1}{2}\left[\frac{j}{2}\delta\left(\frac{1}{2}(f+\frac{1}{\pi})\right) - \frac{j}{2}\delta\left(\frac{1}{2}(f-\frac{1}{\pi})\right)\right] e^{-j\pi f}$$

$$= \left[\frac{j}{2}\delta\left(f+\frac{1}{\pi}\right) - \frac{j}{2}\delta\left(f-\frac{1}{\pi}\right)\right] e^{-j\pi f}$$

Exercise 4 (2)

- Therefore:

$$\sin(2t-1) \rightleftharpoons \frac{j}{2}\delta\left(f+\frac{1}{\pi}\right)e^{-j\pi f} - \frac{j}{2}\delta\left(f-\frac{1}{\pi}\right)e^{-j\pi f}$$
$$= \frac{j}{2}e^{j}\delta\left(f+\frac{1}{\pi}\right) - \frac{j}{2}e^{-j}\delta\left(f-\frac{1}{\pi}\right)$$

Second Method:

- $x(2t-1) = \sin(2t-1)$ can be written as:

$$\sin(2t-1) = \frac{e^{j(2t-1)} - e^{-j(2t-1)}}{2j}$$
$$= -\frac{j}{2}e^{-j}e^{j2\pi\frac{1}{\pi}t} + \frac{j}{2}e^{j}e^{-j2\pi\frac{1}{\pi}t}$$
$$\Rightarrow \quad \mathcal{F}[\sin(2t-1)] = -\frac{j}{2}e^{-j}\delta\left(f-\frac{1}{\pi}\right) + \frac{j}{2}e^{j}\delta\left(f+\frac{1}{\pi}\right)$$

Exercise 5

Determine the Fourier transform of $\sin(2(t-1))$.

$$\sin(t) \rightleftharpoons \frac{j}{2}\delta\left(f+\frac{1}{2\pi}\right) - \frac{j}{2}\delta\left(f-\frac{1}{2\pi}\right)$$

$$t \to 2t \to \sin(2t) \rightleftharpoons \frac{1}{2}\left[\frac{j}{2}\delta\left(\frac{f}{2}+\frac{1}{2\pi}\right) - \frac{j}{2}\delta\left(\frac{f}{2}-\frac{1}{2\pi}\right)\right]$$

$$t \to t-1 \to \sin(2(t-1)) \rightleftharpoons \frac{1}{2}\left[\frac{j}{2}\delta\left(\frac{1}{2}(f+\frac{1}{\pi})\right) - \frac{j}{2}\delta\left(\frac{1}{2}(f-\frac{1}{\pi})\right)\right]e^{-j2\pi f}$$

$$= \frac{j}{2}\delta\left(f+\frac{1}{\pi}\right)e^{-j2\pi f} - \frac{j}{2}\delta\left(f-\frac{1}{\pi}\right)e^{-j2\pi f}$$

$$= \frac{j}{2}e^{2j}\delta\left(f+\frac{1}{\pi}\right) - \frac{j}{2}e^{-2j}\delta\left(f-\frac{1}{\pi}\right)$$

Exercise 6 (1)

Find $x(t)$ when: $x(t) = \mathrm{rect}\,(t) * \cos(\pi t)$.

- *Step 1:* Determine the Fourier transform of $\mathrm{rect}\,(t)$:

$$\mathrm{rect}\,(t) \rightleftharpoons \mathrm{sinc}\,(f)$$

- *Step 2:* Determine the Fourier transform of $\cos(\pi t)$:

$$\begin{aligned}\cos(\pi t) &= \cos\left(2\pi\left(\frac{1}{2}\right)t\right)\\ &\rightleftharpoons \frac{1}{2}\delta\left(f-\frac{1}{2}\right)+\frac{1}{2}\delta\left(f+\frac{1}{2}\right)\end{aligned}$$

- *Step 3:* Determine the Fourier transform of $x\,(t)$:

$$\begin{aligned}X(f) &= \frac{1}{2}\mathrm{sinc}\,(f)\,\delta\left(f-\frac{1}{2}\right)+\frac{1}{2}\mathrm{sinc}\,(f)\,\delta\left(f+\frac{1}{2}\right)\\ &= \frac{1}{2}\mathrm{sinc}\left(\frac{1}{2}\right)\delta\left(f-\frac{1}{2}\right)+\frac{1}{2}\mathrm{sinc}\left(-\frac{1}{2}\right)\delta\left(f+\frac{1}{2}\right)\end{aligned}$$

Exercise 6 (2)

- On the other hand:

$$\operatorname{sinc}\left(\frac{1}{2}\right) = \operatorname{sinc}\left(-\frac{1}{2}\right) = \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = \frac{2}{\pi}$$

- Consequently:

$$X(f) = \frac{2}{\pi}\left[\frac{1}{2}\delta\left(f - \frac{1}{2}\right) + \frac{1}{2}\delta\left(f + \frac{1}{2}\right)\right]$$

$$\Rightarrow x(t) = \mathcal{F}^{-1}\left[X(f)\right] = \frac{2}{\pi}\cos(\pi t)$$

Exercise 7 (1)

Find $x(t)$ when: $x(t) = \mathrm{sinc}\,(t) * \mathrm{sinc}^2\left(\frac{t}{2}\right)$.

- *Step 1:* Determine the Fourier transform of $\mathrm{sinc}\,(t)$:

$$\mathrm{sinc}\,(t) \rightleftharpoons \mathrm{rect}\,(f)$$

- *Step 2:* Determine the Fourier transform of $\mathrm{sinc}^2\left(\frac{t}{2}\right)$:

$$\mathrm{sinc}^2\,(t) \rightleftharpoons \mathrm{tri}(f)$$

$$\mathrm{sinc}^2\left(\frac{t}{2}\right) \rightleftharpoons 2\mathrm{tri}(2f)$$

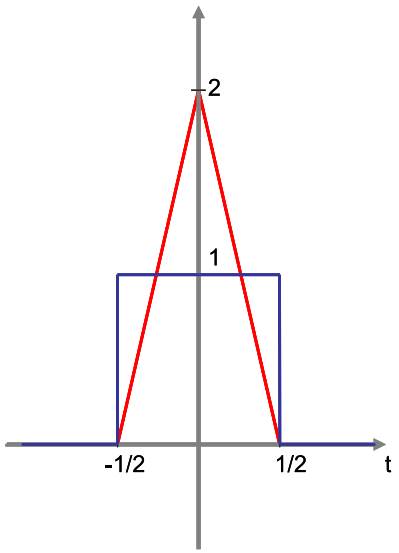
- *Step 3:* Determine the Fourier transform of $x\,(t)$:

$$X(f) = \mathrm{rect}(f)2\mathrm{tri}(2f)$$

Exercise 7 (2)

- On the other hand:

$$X(f) = \mathrm{rect}(f)2\mathrm{tri}(2f) = 2\mathrm{tri}(2f)$$



- Consequently:

$$x(t) = \mathcal{F}^{-1}\left[X(f)\right] = \mathcal{F}^{-1}\left[2\mathrm{tri}(2f)\right] = \mathrm{sinc}^2\left(\frac{t}{2}\right)$$

Exercise 8 (1)

- $X(f)$ can be determined in two methods.

First Method:

- $x(t)$ can be written as:

$$x(t) = \frac{1}{2}e^{j200\pi t}e^{-j\frac{\pi}{4}} + \frac{1}{2}e^{-j200\pi t}e^{j\frac{\pi}{4}}$$

$$\Rightarrow \; X(f) = \frac{1}{2}e^{-j\frac{\pi}{4}}\delta(f-100) + \frac{1}{2}e^{j\frac{\pi}{4}}\delta(f+100)$$

Second Method:

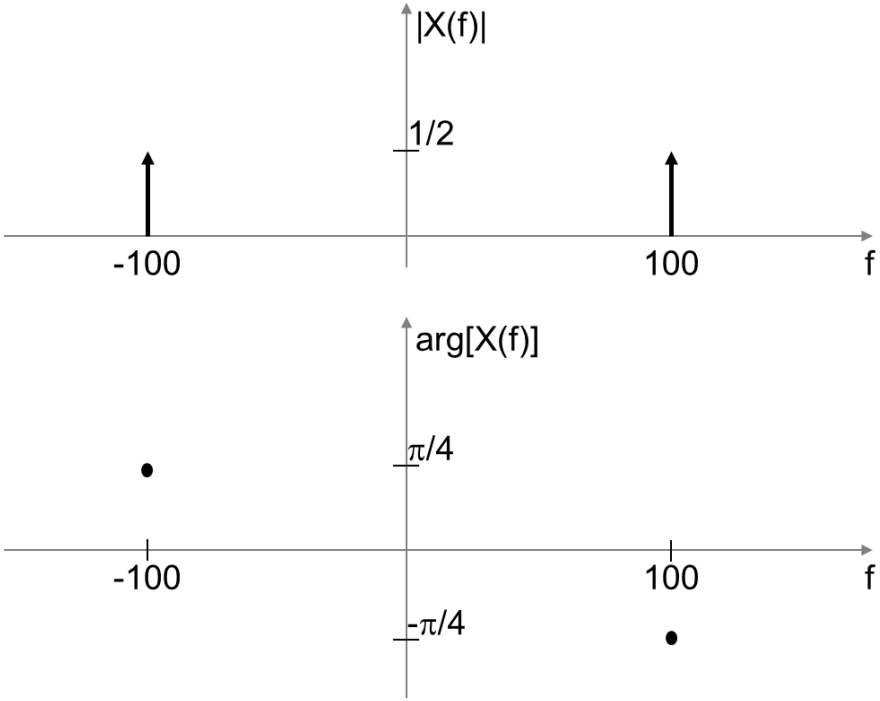
- $x(t)$ can be written as:

$$x(t) = \cos\left(200\pi\left(t - \frac{1}{800}\right)\right)$$

$$\begin{aligned}
\Rightarrow \quad X(f) &= \left[\frac{1}{2}\delta(f-100) + \frac{1}{2}\delta(f+100)\right]e^{-j\frac{2\pi}{800}f} \\
&= \frac{1}{2}e^{-j\frac{2\pi}{800}100}\delta(f-100) + \frac{1}{2}e^{-j\frac{2\pi}{800}(-100)}\delta(f+100) \\
&= \frac{1}{2}e^{-j\frac{\pi}{4}}\delta(f-100) + \frac{1}{2}e^{j\frac{\pi}{4}}\delta(f+100)
\end{aligned}$$

Exercise 8 (2)

- The magnitude and phase spectra are shown below:



  - Note that the signal is real implying that the magnitude spectrum is even while the phase spectrum is odd.

Exercise 9 (a.1)

First Method:

- *Step 1:* Calculate the Fourier transform of $x(t) = 4\text{sinc}\left(\frac{t}{5}\right)$:

$$\text{sinc}(t) \rightleftharpoons \text{rect}(f)$$
$$4\text{sinc}(t) \rightleftharpoons 4\text{rect}(f)$$
$$t \to \frac{t}{5} \Rightarrow 4\text{sinc}\left(\frac{t}{5}\right) \rightleftharpoons 4\left[5\text{rect}(5f)\right] = 20\text{rect}(5f)$$

- *Step 2:* Apply Parseval's theorem:

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 \mathrm{d}t = \int_{-\infty}^{+\infty} |X(f)|^2 \mathrm{d}f$$
$$= \int_{-\infty}^{+\infty} 400\text{rect}^2(5f)\mathrm{d}f = \int_{-\frac{1}{10}}^{+\frac{1}{10}} 400\mathrm{d}f = 400\frac{2}{10} = 80 \ \ \mathrm{J}$$

Exercise 9 (a.2)

Second Method:

- Let $y(t) = x^2(t) = 16\mathrm{sinc}^2\left(\frac{t}{5}\right)$.
- The energy of $x(t)$ is given by:

$$E = \int_{-\infty}^{+\infty} x^2(t)\mathrm{d}t = \int_{-\infty}^{+\infty} y(t)\mathrm{d}t = Y(0)$$

  where the last equality follows from the total area theorem.
- On the other hand:

$$Y(f) = 16 \times 5\mathrm{tri}(5f) = 80\mathrm{tri}(5f)$$

- Consequently:

$$E = Y(0) = 80\underbrace{\mathrm{tri}(0)}_{=1} = 80 \ \ \mathrm{J}$$

Exercise 9 (b.1)

- The FT of $y(t) = x(t-1) = 4\text{sinc}\left(\frac{t-1}{5}\right)$ is:

$$Y(f) = X(f)e^{-j2\pi f} = 20\text{rect}\,(5f)\,e^{-j2\pi f}$$

- Applying Parseval's theorem, the energy of $y(t)$ is given by:

$$E = \int_{-\infty}^{+\infty} |y(t)|^2 \mathrm{d}t = \int_{-\infty}^{+\infty} |Y(f)|^2 \mathrm{d}f$$
$$= \int_{-\infty}^{+\infty} 400\text{rect}^2(5f)\mathrm{d}f = \int_{-\frac{1}{10}}^{+\frac{1}{10}} 400\mathrm{d}f = 400\frac{2}{10} = 80 \ \ \mathrm{J}$$

- Note that $|Y(f)| = |20\text{rect}\,(5f)\,e^{-j2\pi f}| = 20\text{rect}(5f)$ since $|e^{-j2\pi f}| = 1$.
- Note that $x(t)$ and $x(t-1)$ have the same energy; i.e. time-shifting does not change the energy of the signal. In fact, the area under the magnitude-squared of the signal will remain the same.

Exercise 10

Using Parseval's theorem calculate the energy of $x(t) = 2\mathrm{sinc}^2(3t)$

- *Step 1:* Calculate the Fourier transform of $2\mathrm{sinc}^2(3t)$:

$$\mathrm{sinc}^2(t) \rightleftharpoons \mathrm{tri}(f)$$

$$2\mathrm{sinc}^2(t) \rightleftharpoons 2\mathrm{tri}(f)$$

$$t \to 3t \Rightarrow 2\mathrm{sinc}^2(3t) \rightleftharpoons 2\left[\frac{1}{3}\mathrm{tri}\left(\frac{f}{3}\right)\right] = \frac{2}{3}\mathrm{tri}\left(\frac{f}{3}\right)$$

- *Step 2:* Apply Parseval's theorem:

$$E = \int_{-\infty}^{+\infty} |x(t)|^2 \mathrm{d}t = \int_{-\infty}^{+\infty} |X(f)|^2 \mathrm{d}f = \frac{4}{9}\int_{-\infty}^{+\infty} \mathrm{tri}^2\left(\frac{f}{3}\right)\mathrm{d}f$$

$$= \frac{4}{9}\int_{-3}^{+3} \mathrm{tri}^2\left(\frac{f}{3}\right)\mathrm{d}f = \frac{8}{9}\int_{0}^{+3} \mathrm{tri}^2\left(\frac{f}{3}\right)\mathrm{d}f$$

$$= \frac{8}{9}\int_{0}^{+3} \left(-\frac{f}{3}+1\right)^2 \mathrm{d}f = \cdots = \frac{8}{9}\ \mathrm{J}$$

Exercise 11 (1)

First Method:

- $y(t)$ can be written as:

$$\begin{aligned} y(t) &= x(2t) \\ &= \cos(200\pi t) + 2\sin(300\pi t) \\ &= \cos(2\pi(100)t) + 2\sin(2\pi(150)t) \end{aligned}$$

implying that:

$$Y(f) = \frac{1}{2}\delta(f-100) + \frac{1}{2}\delta(f+100) + j\delta(f+150) - j\delta(f-150)$$

Exercise 11 (2)

Second Method:

- We first calculate the FT of x(t):

$$x(t) = \cos(100\pi t) + 2\sin(150\pi t) = \cos(2\pi(50)t) + 2\sin(2\pi(75)t)$$

implying that:

$$X(f) = \frac{1}{2}\delta(f-50) + \frac{1}{2}\delta(f+50) + j\delta(f+75) - j\delta(f-75)$$

- The FT of $y(t) = x(2t)$ is $Y(f) = \frac{1}{2}X\left(\frac{f}{2}\right)$:

$$Y(f) = \frac{1}{4}\delta\left(\frac{f}{2}-50\right) + \frac{1}{4}\delta\left(\frac{f}{2}+50\right) + \frac{j}{2}\delta\left(\frac{f}{2}+75\right) - \frac{j}{2}\delta\left(\frac{f}{2}-75\right)$$

$$= \frac{1}{4}\delta\left(\frac{1}{2}(f-100)\right) + \frac{1}{4}\delta\left(\frac{1}{2}(f+100)\right) + \frac{j}{2}\delta\left(\frac{1}{2}(f+150)\right) - \frac{j}{2}\delta\left(\frac{1}{2}(f-150)\right)$$

$$= \frac{1}{2}\delta(f-100) + \frac{1}{2}\delta(f+100) + j\delta(f+150) - j\delta(f-150)$$

where the relation $\delta(a(f-f_0)) = \frac{1}{|a|}\delta(f-f_0)$ was applied.

Exercise 12 (a)

- $\sin(2\pi f_0 t)$ is related to $\cos(2\pi f_0 t)$ by:

$$\sin(2\pi f_0 t) = \cos\left(2\pi f_0 t - \frac{\pi}{2}\right) = \cos\left(2\pi f_0\left(t - \frac{\pi}{4\pi f_0}\right)\right)$$

$$\begin{aligned}\Rightarrow \quad \mathcal{F}[\sin(2\pi\, f_0 t)] &= \mathcal{F}\left[\cos\left(2\pi f_0\left(t - \frac{\pi}{4\pi f_0}\right)\right)\right] \\ &= \left[\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)\right] e^{-j2\pi \frac{\pi}{4\pi f_0} f}\end{aligned}$$

- Consequently:

$$\begin{aligned}\mathcal{F}[\sin(2\pi\, f_0 t)] &= \frac{1}{2}\left[e^{-j\frac{\pi}{2f_0}f}\right]_{f=f_0}\delta(f - f_0) + \frac{1}{2}\left[e^{-j\frac{\pi}{2f_0}f}\right]_{f=-f_0}\delta(f + f_0) \\ &= \frac{1}{2}e^{-j\frac{\pi}{2}}\delta(f - f_0) + \frac{1}{2}e^{j\frac{\pi}{2}}\delta(f + f_0) \\ &= -\frac{j}{2}\delta(f - f_0) + \frac{j}{2}\delta(f + f_0)\end{aligned}$$

Exercise 12 (b)

- $\sin(2\pi f_0 t)$ is related to $\cos(2\pi f_0 t)$ by:

$$\sin(2\pi f_0 t) = -\frac{1}{2\pi f_0}\frac{\mathrm{d}[\cos(2\pi f_0 t)]}{\mathrm{d}t}$$

- Taking the Fourier transform of both sides:

$$\begin{aligned}
\mathcal{F}[\sin(2\pi f_0 t)] &= -\frac{1}{2\pi f_0}\mathcal{F}\left[\frac{\mathrm{d}[\cos(2\pi f_0 t)]}{\mathrm{d}t}\right] \\
&= -\frac{1}{2\pi f_0}\left[(j2\pi f)\mathcal{F}[\cos(2\pi f_0 t)]\right] \\
&= -\frac{jf}{f_0}\left[\frac{1}{2}\delta(f-f_0)+\frac{1}{2}\delta(f+f_0)\right] \\
&= \left[-\frac{jf}{2f_0}\right]_{f=f_0}\delta(f-f_0)+\left[-\frac{jf}{2f_0}\right]_{f=-f_0}\delta(f+f_0) \\
&= -\frac{j}{2}\delta(f-f_0)+\frac{j}{2}\delta(f+f_0)
\end{aligned}$$

Exercise 13

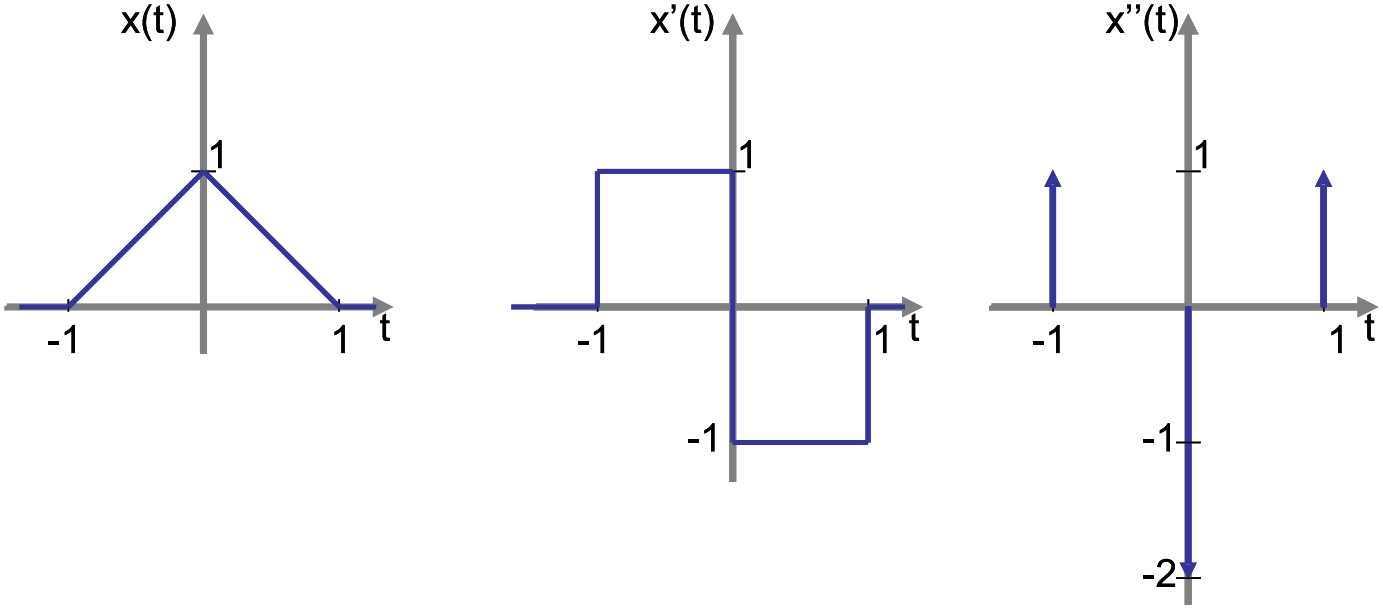
Find the Fourier transform of $x(t) = \delta(t+5) + 2\delta(t) + \delta(t-5)$.

$$\begin{aligned} X(f) = \mathcal{F}[x(t)] &= e^{j2\pi 5f} + 2 + e^{-j2\pi 5f} \\ &= 2 + e^{j10\pi f} + e^{-j10\pi f} \\ &= 2 + 2\cos(10\pi f) \end{aligned}$$

- $x(t)$ is real and even.
- $X(f)$ is real and even.

Exercise 14 (1)

Plot $x(t)$, $x'(t)$ and $x''(t)$.



Exercise 14 (2)

Calculate the Fourier transform of $x(t)$, $x'(t)$ and $x''(t)$.

$$x(t) = \mathrm{tri}\,(t) \rightleftharpoons X(f) = \mathrm{sinc}^2(f)$$

$$x'(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} \rightleftharpoons X'(f) = (j2\pi f)X(f)$$

$$x''(t) = \frac{\mathrm{d}x'(t)}{\mathrm{d}t} \rightleftharpoons X''(f) = (j2\pi f)X'(f) = -4\pi^2 f^2 X(f)$$

- This implies that:

$$X''(f) = -4\pi^2 f^2 \frac{\sin^2(\pi f)}{\pi^2 f^2} = -4\sin^2(\pi f) = 2\left[\cos(2\pi f) - 1\right]$$

- Calculating the inverse Fourier transform of $X''(f)$:

$$\begin{aligned} x''(t) &= 2\left[\frac{1}{2}\delta(t-1) + \frac{1}{2}\delta(t+1) - \delta(t)\right] \\ &= \delta(t-1) + \delta(t+1) - 2\delta(t) \end{aligned}$$

Exercise 15 (1)

$$H(f) = \frac{5}{1 + j\pi f}$$

- $x(t) = 3\delta(t)$:
  - $Y(f)$ is given by:

$$Y(f) = X(f)H(f) = 3\frac{5}{1 + j\pi f} = \frac{15}{1 + j\pi f}$$

  - We have previously proven that:

$$e^{-at}u(t) \rightleftharpoons \frac{1}{a + j2\pi f}$$

  - Consequently:

$$Y(f) = \frac{15}{1 + j\pi f} = \frac{30}{2 + j2\pi f}$$

$$\Rightarrow \; y(t) = \mathcal{F}^{-1}\left[Y(f)\right] = 30e^{-2t}u(t)$$

Exercise 15 (2)

- $x(t) = \cos(4t)$:
  - Determine $X(f)$:

$$x(t) = \cos\left(2\pi\left(\frac{2}{\pi}\right)t\right) \rightleftharpoons \frac{1}{2}\delta\left(f - \frac{2}{\pi}\right) + \frac{1}{2}\delta\left(f + \frac{2}{\pi}\right)$$

  - Determine $Y(f)$ and $y(t)$:

$$\begin{aligned}
Y(f) &= H(f)X(f) \\
&= \frac{5}{1+j\pi f}\frac{1}{2}\delta\left(f - \frac{2}{\pi}\right) + \frac{5}{1+j\pi f}\frac{1}{2}\delta\left(f + \frac{2}{\pi}\right) \\
&= \frac{5}{2}\left[\frac{1}{1+j\pi\frac{2}{\pi}}\delta\left(f - \frac{2}{\pi}\right) + \frac{1}{1+j\pi\left(-\frac{2}{\pi}\right)}\delta\left(f + \frac{2}{\pi}\right)\right] \\
&= \frac{5}{2}\left[\frac{1}{1+2j}\delta\left(f - \frac{2}{\pi}\right) + \frac{1}{1-2j}\delta\left(f + \frac{2}{\pi}\right)\right] \\
&= \frac{1}{2}\left[\delta\left(f - \frac{2}{\pi}\right) + \delta\left(f + \frac{2}{\pi}\right) + 2j\left[\delta\left(f + \frac{2}{\pi}\right) - \delta\left(f - \frac{2}{\pi}\right)\right]\right]
\end{aligned}$$

$$\Rightarrow y(t) = \cos(4t) + 2\sin(4t)$$

Exercise 15 (3)

- Note that it was possible to solve this problem by writing $\frac{1}{1+2j}$ and $\frac{1}{1-2j}$ in polar form:

$$
\begin{aligned}
Y(f) &= \frac{5}{2}\left[\frac{1}{1+2j}\delta\left(f-\frac{2}{\pi}\right)+\frac{1}{1-2j}\delta\left(f+\frac{2}{\pi}\right)\right] \\
&= \frac{5}{2}\left[\frac{1}{\sqrt{5}}e^{-1.1071j}\delta\left(f-\frac{2}{\pi}\right)+\frac{1}{\sqrt{5}}e^{1.1071j}\delta\left(f+\frac{2}{\pi}\right)\right] \\
\Rightarrow \quad y(t) &= \frac{5}{2}\left[\frac{1}{\sqrt{5}}e^{-1.1071j}e^{j2\pi\frac{2}{\pi}t}+\frac{1}{\sqrt{5}}e^{1.1071j}e^{-j2\pi\frac{2}{\pi}t}\right] \\
&= \frac{\sqrt{5}}{2}\left[e^{j(4t-1.1071)}+e^{-j(4t-1.1071)}\right] \\
&= \sqrt{5}\cos(4t-1.1071)
\end{aligned}
$$

- We can check the result by developing $y(t)$ using $\cos(a-b)=\cos(a)\cos(b)+\sin(a)\sin(b)$ to find that:

$$y(t)=\sqrt{5}\cos(4t-1.1071)=\cos(4t)+2\sin(4t)$$

Exercise 15 (4)

Second Method:

- Since $h(t) = 10e^{-2t}u(t)$, then $h(t)$ is real valued. Consequently:

$$y(t) = |H(f_0)| \cos(2\pi f_0 t + \arg[H(f_0)]) \;\; ; \;\; f_0 = \frac{2}{\pi}$$

- $H(f_0)$ is given by:

$$H(f_0) = \frac{5}{1 + j\pi f_0} = \frac{5}{1 + 2j}$$

  - $|H(f_0)| = \sqrt{5}$.
  - $\arg[H(f_0)] = -\tan^{-1}(2) = -1.1071$ radians.

- Consequently:

$$y(t) = \sqrt{5}\cos(4t - 1.1071)$$

Exercise 15 (5)

- $x(t) = e^{-3t}u(t)$:
  - $X(f)$ is given by:

$$X(f) = \frac{1}{3 + j2\pi f}$$

  - $Y(f)$ is given by:

$$\begin{aligned} Y(f) &= X(f)H(f) \\ &= 10\frac{1}{3 + j2\pi f}\frac{1}{2 + j2\pi f} \end{aligned}$$

  - Determine the values of A and B such that:

$$Y(f) = \frac{A}{3 + j2\pi f} + \frac{B}{2 + j2\pi f}$$

$$A = \left.\frac{10}{2 + j2\pi f}\right]_{j2\pi f = -3} = -10$$

$$B = \left.\frac{10}{3 + j2\pi f}\right]_{j2\pi f = -2} = 10$$

Exercise 15 (6)

$$Y(f) = -\frac{10}{3 + j2\pi f} + \frac{10}{2 + j2\pi f}$$

$$\Rightarrow \; y(t) = -10e^{-3t}u(t) + 10e^{-2t}u(t)$$

Exercise 15 (7)

- $x(t) = e^{-t}\mathrm{rect}(t - 1/2)$ can be written as:

$$x(t) = e^{-t}[u(t) - u(t-1)] = e^{-t}u(t) - e^{-1}e^{-(t-1)}u(t-1)$$

- We first determine the response to $x_1(t) = e^{-t}u(t)$.
  - $X_1(f)$ is given by:

$$X(f) = \frac{1}{1 + j2\pi f}$$

  - $Y_1(f)$ is given by:

$$Y_1(f) = X_1(f)H(f) = 10\frac{1}{1 + j2\pi f}\frac{1}{2 + j2\pi f}$$

  - Determine the values of A and B such that:

$$Y_1(f) = \frac{A}{1 + j2\pi f} + \frac{B}{2 + j2\pi f}$$

$$A = \left.\frac{10}{2 + j2\pi f}\right]_{j2\pi f = -1} = 10$$

$$B = \left.\frac{10}{1 + j2\pi f}\right]_{j2\pi f = -2} = -10$$

Exercise 15 (8)

- Consequently:

$$Y_1(f) = \frac{10}{1 + j2\pi f} - \frac{10}{2 + j2\pi f}$$
$$\Rightarrow \; y_1(t) = 10e^{-t}u(t) - 10e^{-2t}u(t) = 10(e^{-t} - e^{-2t})u(t)$$

- Since the system is LTI, then the response to $x(t)$ is given by:

$$\begin{aligned} y(t) &= y_1(t) - e^{-1}y_1(t-1) \\ &= 10(e^{-t} - e^{-2t})u(t) - e^{-1}10(e^{-(t-1)} - e^{-2(t-1)})u(t-1) \end{aligned}$$

Exercise 16 (1)

$$X(f) = \frac{9}{(2+j2\pi f)(1+j5\pi f)} + \delta(f)$$

- Determine the Fourier transform of $x(-3t)$:

$$x(t) \rightleftharpoons X(f)$$

$$t \to -t \Rightarrow x(-t) \rightleftharpoons X(-f) = \frac{9}{(2-j2\pi f)(1-j5\pi f)} + \delta(-f)$$

$$= \frac{9}{(2-j2\pi f)(1-j5\pi f)} + \delta(f)$$

$$t \to 3t \Rightarrow x(-3t) \rightleftharpoons \frac{1}{3}X\left(-\frac{f}{3}\right) = \frac{1}{3}\frac{9}{(2-j2\pi\frac{f}{3})(1-j5\pi\frac{f}{3})} + \frac{1}{3}\delta\left(\frac{f}{3}\right)$$

$$= \frac{27}{(6-j2\pi f)(3-j5\pi f)} + \delta(f)$$

Exercise 16 (2)

$$X(f) = \frac{9}{(2+j2\pi f)(1+j5\pi f)} + \delta(f)$$

- Determine the Fourier transform of $x(3t-2)$:

$$t \to t-2 \Rightarrow x(t-2) \rightleftharpoons X(f)e^{j2\pi(-2)f}$$

$$\begin{aligned}
t \to 3t \Rightarrow x(3t-2) &\rightleftharpoons \frac{1}{3}X\left(\frac{f}{3}\right)e^{-j4\pi\frac{f}{3}} \\
&= \frac{1}{3}\left[\frac{9}{(2+j2\pi\frac{f}{3})(1+j5\pi\frac{f}{3})}e^{-j4\pi\frac{f}{3}} + \delta(\frac{f}{3})e^{-j4\pi\frac{f}{3}}\right] \\
&= \frac{27}{(6+j2\pi f)(3+j5\pi f)}e^{-j\frac{4}{3}\pi f} + \delta(f)e^{-j4\pi\frac{f}{3}} \\
&= \frac{27}{(6+j2\pi f)(3+j5\pi f)}e^{-j\frac{4}{3}\pi f} + \delta(f)
\end{aligned}$$

Exercise 16 (3)

$$X(f) = \frac{9}{(2 + j2\pi f)(1 + j5\pi f)} + \delta(f)$$

- Determine the Fourier transform of $x(t)\cos(4t)$:

$$\cos(4t) = \cos\left(2\pi\left(\frac{2}{\pi}\right)t\right) \rightleftharpoons \frac{1}{2}\delta\left(f + \frac{2}{\pi}\right) + \frac{1}{2}\delta\left(f - \frac{2}{\pi}\right)$$

$$x(t)\cos(4t) \rightleftharpoons \frac{1}{2}X\left(f + \frac{2}{\pi}\right) + \frac{1}{2}X\left(f - \frac{2}{\pi}\right)$$

- Therefore:

$$x(t)\cos(4t) \rightleftharpoons \frac{1}{2}\left[\frac{9}{(2 + j2\pi(f + \frac{2}{\pi}))(1 + j5\pi(f + \frac{2}{\pi}))} + \frac{9}{(2 + j2\pi(f - \frac{2}{\pi}))(1 + j5\pi(f - \frac{2}{\pi}))} + \delta\left(f + \frac{2}{\pi}\right) + \delta\left(f - \frac{2}{\pi}\right)\right]$$

Exercise 16 (4)

$$X(f) = \frac{9}{(2 + j2\pi f)(1 + j5\pi f)} + \delta(f)$$

- Determine the Fourier transform of $\frac{\mathrm{d}x(t)}{\mathrm{d}t}$:

$$\begin{aligned}
x(t) &\rightleftharpoons X(f) \\
\frac{\mathrm{d}x(t)}{\mathrm{d}t} &\rightleftharpoons j2\pi f X(f) \\
&= \frac{18j\pi f}{(2 + j2\pi f)(1 + j5\pi f)} + j2\pi f\,\delta(f) \\
&= \frac{18j\pi f}{(2 + j2\pi f)(1 + j5\pi f)} + [j2\pi f]_{f=0}\,\delta(f) \\
&= \frac{18j\pi f}{(2 + j2\pi f)(1 + j5\pi f)}
\end{aligned}$$

Exercise 17 (1)

- The FT of $x(t)$ is given by:

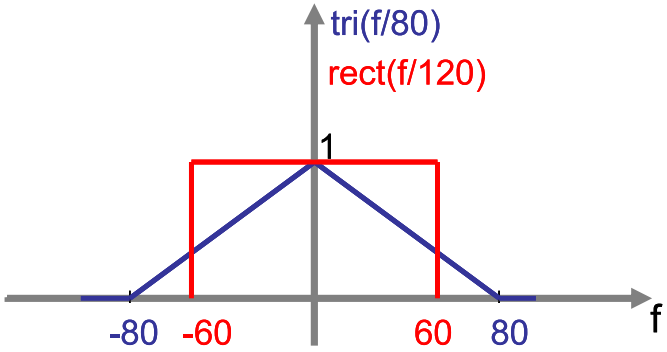
$$X(f) = 10\delta(f) + 2\delta(f-50) + 2\delta(f+50) + j\delta(f+100) - j\delta(f-100) + \mathrm{tri}\left(\frac{f}{80}\right)$$

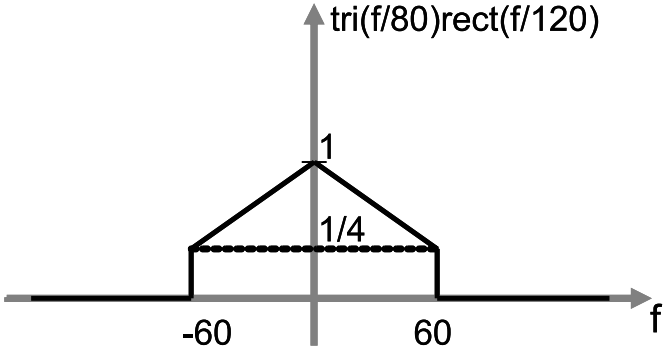
- At the output of the filter having a cutoff-frequency of 60 Hz $H(f) = \mathrm{rect}\left(\frac{f}{120}\right)$:
  - The impulses located at the frequencies 0, +50 and -50 will pass through the filter.
  - The impulses located at the frequencies +100 and -100 will be eliminated by the filter.
- Consequently, the FT of the output signal is given by:

$$Y(f) = 10\delta(f) + 2\delta(f-50) + 2\delta(f+50) + \mathrm{tri}\left(\frac{f}{80}\right)\mathrm{rect}\left(\frac{f}{120}\right)$$

Exercise 17 (2)

- The term $\mathrm{tri}\left(\frac{f}{80}\right)\mathrm{rect}\left(\frac{f}{120}\right)$ is plotted in the figure below:





  - Note that for $f \geq 0$, $\mathrm{tri}\left(\frac{f}{80}\right) = -\frac{f}{80} + 1$ which is equal to 1/4 for $f = 60$.

- Consequently:

$$\mathrm{tri}\left(\frac{f}{80}\right)\mathrm{rect}\left(\frac{f}{120}\right) = \frac{1}{4}\mathrm{rect}\left(\frac{f}{120}\right) + \frac{3}{4}\mathrm{tri}\left(\frac{f}{60}\right)$$

  and its inverse FT is given by: $30\mathrm{sinc}(120t) + 45\mathrm{sinc}^2(60t)$.

- Finally:

$$y(t) = 10 + 4\cos(100\pi t) + 30\mathrm{sinc}(120t) + 45\mathrm{sinc}^2(60t)$$

Exercise 18 (1)

- The FT of the input signal $x(t)$ is given by:

$$
\begin{aligned}
X(f) &= 0.5\delta(f) + 0.75\delta\left(f - \frac{1}{3}\right) + 0.75\delta\left(f + \frac{1}{3}\right) \\
&\qquad + 0.25j\delta\left(f + \frac{1}{3}\right) - 0.25j\delta\left(f - \frac{1}{3}\right) \\
&= 0.5\delta(f) + (0.75 - 0.25j)\delta\left(f - \frac{1}{3}\right) + (0.75 + 0.25j)\delta\left(f + \frac{1}{3}\right)
\end{aligned}
$$

- Consequently, the FT of the output signal $y(t)$ is given by:

$$
\begin{aligned}
Y(f) &= H(f)X(f) \\
&= \frac{0.5}{1 + j2\pi f}\delta(f) + \frac{0.75 - 0.25j}{1 + j2\pi f}\delta\left(f - \frac{1}{3}\right) + \frac{0.75 + 0.25j}{1 + j2\pi f}\delta\left(f + \frac{1}{3}\right)
\end{aligned}
$$

Exercise 18 (2)

- Following from the multiplication property of the impulse:

$$Y(f) = \left[\frac{0.5}{1+j2\pi f}\right]_{f=0} \delta(f) + \left[\frac{0.75-0.25j}{1+j2\pi f}\right]_{f=\frac{1}{3}} \delta\left(f-\frac{1}{3}\right) + \left[\frac{0.75+0.25j}{1+j2\pi f}\right]_{f=-\frac{1}{3}} \delta\left(f+\frac{1}{3}\right)$$

$$= 0.5\delta(f) + \frac{0.75-0.25j}{1+j2\pi/3}\delta\left(f-\frac{1}{3}\right) + \frac{0.75+0.25j}{1-j2\pi/3}\delta\left(f+\frac{1}{3}\right)$$

$$= 0.5\delta(f) + 0.34e^{-j1.44}\delta\left(f-\frac{1}{3}\right) + 0.34e^{j1.44}\delta\left(f+\frac{1}{3}\right)$$

- Consequently, $y(t)$ is given by:

$$y(t) = 0.5 + 0.34e^{-j1.44}e^{j\frac{2\pi}{3}t} + 0.34e^{j1.44}e^{-j\frac{2\pi}{3}t}$$

$$= 0.5 + 0.34\left[e^{j\left(\frac{2\pi}{3}t-1.44\right)} + e^{-j\left(\frac{2\pi}{3}t-1.44\right)}\right]$$

$$= 0.5 + 0.68\cos\left(\frac{2\pi}{3}t - 1.44\right)$$

Exercise 19 (a)

- $x(t)$ can be written as:

$$x(t) = \mathrm{tri}(t+1) - \mathrm{tri}(t-1)$$

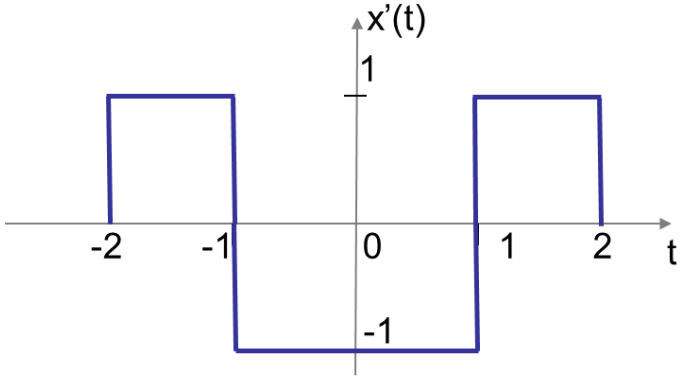
implying that:

$$\begin{aligned} X(f) &= \mathrm{sinc}^2(f)e^{j2\pi f} - \mathrm{sinc}^2(f)e^{-j2\pi f} \\ &= \mathrm{sinc}^2(f)\left[e^{j2\pi f} - e^{-j2\pi f}\right] \\ &= 2j\mathrm{sinc}^2(f)\sin(2\pi f) \end{aligned}$$

  - Note that $x(t)$ is real and odd while $X(f)$ is pure-imaginary and odd.

Exercise 19 (b.1)

- By differentiating $x(t)$, $x'(t)$ is as shown in the figure below:



implying that:

$$x'(t) = \text{rect}\left(t + \frac{3}{2}\right) - \text{rect}\left(\frac{t}{2}\right) + \text{rect}\left(t - \frac{3}{2}\right)$$

- Consequently, the FT of $x'(t)$ can be written as:

$$\begin{aligned}\mathcal{F}\left[x'(t)\right] &= \text{sinc}(f)e^{j2\pi\frac{3}{2}f} - 2\text{sinc}(2f) + \text{sinc}(f)e^{-j2\pi\frac{3}{2}f} \\ &= 2\text{sinc}(f)\cos(3\pi f) - 2\text{sinc}(2f)\end{aligned}$$

Exercise 19 (b.2)

- We next apply the integration property of the FT:

$$\int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \rightleftharpoons \frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$$

$$\Rightarrow \quad x(t) = \int_{-\infty}^{t} x'(\tau)\mathrm{d}\tau \rightleftharpoons X(f) = \frac{\mathcal{F}\left[x'(t)\right]}{j2\pi f} + \frac{1}{2}\left[\mathcal{F}\left[x'(t)\right]\right]_{f=0}\delta(f)$$

- Consequently:

$$X(f) = \frac{2\mathrm{sinc}(f)\cos(3\pi f) - 2\mathrm{sinc}(2f)}{j2\pi f} + \frac{1}{2} \times 0 \times \delta(f)$$

$$= \frac{1}{j\pi f}\left[\mathrm{sinc}(f)\cos(3\pi f) - \mathrm{sinc}(2f)\right]$$

Exercise 19 (c)

- To prove that the expressions in (a) and (b) are equal, the last expression of $X(f)$ can be written as:

$$
\begin{aligned}
X(f) &= \frac{1}{j\pi f}\left[\frac{\sin(\pi f)\cos(3\pi f)}{\pi f} - \frac{\sin(2\pi f)}{2\pi f}\right] \\
&= \frac{1}{j\pi f}\left[\frac{\sin(\pi f)\cos(3\pi f)}{\pi f} - \frac{2\sin(\pi f)\cos(\pi f)}{2\pi f}\right] \\
&= \frac{1}{j(\pi f)^2}\sin(\pi f)\left[\cos(3\pi f) - \cos(\pi f)\right] \\
&= -\frac{2}{j(\pi f)^2}\sin(\pi f)\left[\sin(\pi f)\sin(2\pi f)\right]
\end{aligned}
$$

  - where the expression $\cos(a) - \cos(b) = -2\sin\left(\frac{a-b}{2}\right)\sin\left(\frac{a+b}{2}\right)$ was used.

- Finally:

$$X(f) = 2j\,\mathrm{sinc}^2(f)\sin(2\pi f)$$

which is the same as the expression obtained in part (a).

Exercise 20 (a)

- Using the definition:

$$X(f) = \int_{-\infty}^{+\infty} e^{-at}\sin(bt)u(t)e^{-j2\pi ft}\mathrm{d}t = \int_{0}^{+\infty} e^{-at}\frac{e^{jbt}-e^{-jbt}}{2j}e^{-j2\pi ft}\mathrm{d}t$$

$$= \frac{1}{2j}\int_{0}^{+\infty} e^{-[a-j(b-2\pi f)]t}\mathrm{d}f - \frac{1}{2j}\int_{0}^{+\infty} e^{-[a+j(b+2\pi f)]t}\mathrm{d}f$$

$$= -\frac{1}{2j}\frac{1}{a-j(b-2\pi f)}\underbrace{\left[e^{-[a-j(b-2\pi f)]t}\right]_0^{+\infty}}_{=-1}$$

$$+\frac{1}{2j}\frac{1}{a+j(b+2\pi f)}\underbrace{\left[e^{-[a+j(b+2\pi f)]t}\right]_0^{+\infty}}_{=-1}$$

$$= \frac{1}{2j}\left[\frac{1}{a-j(b-2\pi f)} - \frac{1}{a+j(b+2\pi f)}\right]$$

Exercise 20 (b)

- Using the multiplication-convolution duality:

$$
\begin{aligned}
x(t) &= \left[e^{-at}u(t)\right]\left[\sin\left(2\pi\frac{b}{2\pi}t\right)\right] \\
\Rightarrow \quad X(f) &= \left[\frac{1}{a+j2\pi f}\right] * \left[\frac{j}{2}\delta\left(f+\frac{b}{2\pi}\right) - \frac{j}{2}\delta\left(f-\frac{b}{2\pi}\right)\right] \\
&= \frac{j}{2}\frac{1}{a+j2\pi\left(f+\frac{b}{2\pi}\right)} - \frac{j}{2}\frac{1}{a+j2\pi\left(f-\frac{b}{2\pi}\right)} \\
&= \frac{j}{2}\frac{1}{a+j\left(2\pi f+b\right)} - \frac{j}{2}\frac{1}{a+j\left(2\pi f-b\right)} \\
&= \frac{1}{2j}\left[\frac{1}{a-j(b-2\pi f)} - \frac{1}{a+j(b+2\pi f)}\right]
\end{aligned}
$$

which is the same as the result in part (a).

Exercise 21 (1)

- The Fourier transforms of $x_1(t)$ and $x_2(t)$ are given by:

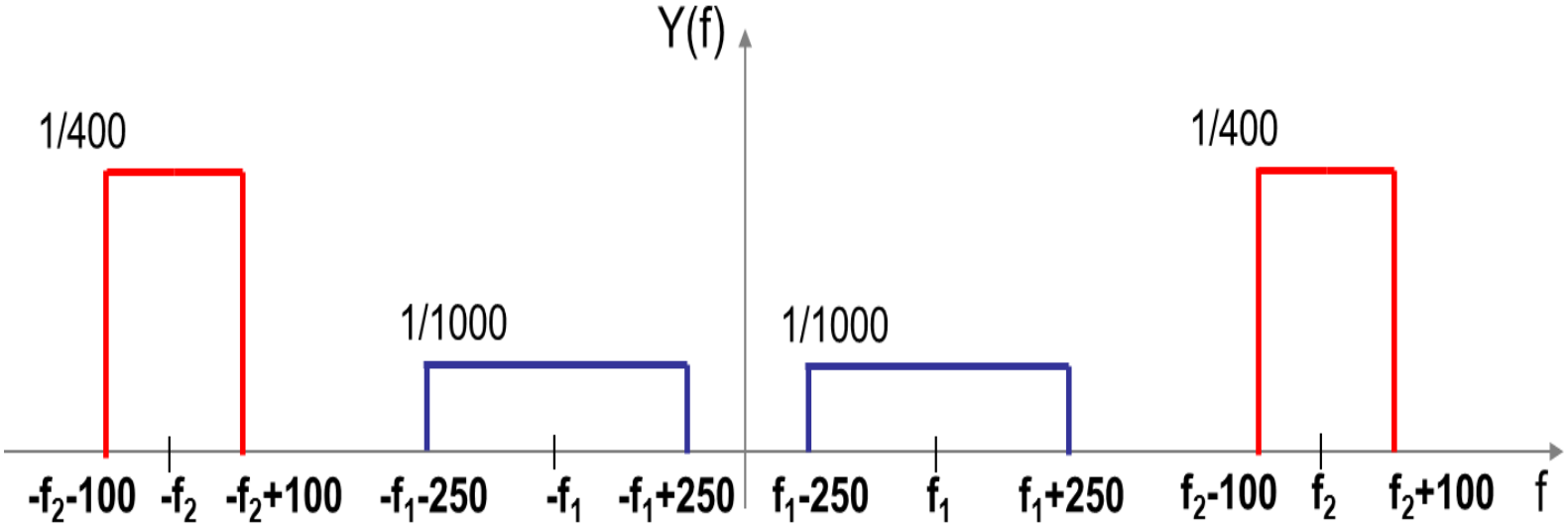
$$X_1(f) = \frac{1}{500}\text{rect}\left(\frac{f}{500}\right) \quad ; \quad X_2(f) = \frac{1}{200}\text{rect}\left(\frac{f}{200}\right)$$

- The Fourier transform of the FDM signal $y(t)$ is:

$$\begin{aligned} Y(f) &= \frac{1}{2}X_1(f-f_1) + \frac{1}{2}X_1(f+f_1) + \frac{1}{2}X_2(f-f_2) + \frac{1}{2}X_2(f+f_2) \\ &= \frac{1}{1000}\text{rect}\left(\frac{f-f_1}{500}\right) + \frac{1}{1000}\text{rect}\left(\frac{f+f_1}{500}\right) \\ &\quad + \frac{1}{400}\text{rect}\left(\frac{f-f_2}{200}\right) + \frac{1}{400}\text{rect}\left(\frac{f+f_2}{200}\right) \end{aligned}$$

Exercise 21 (2)

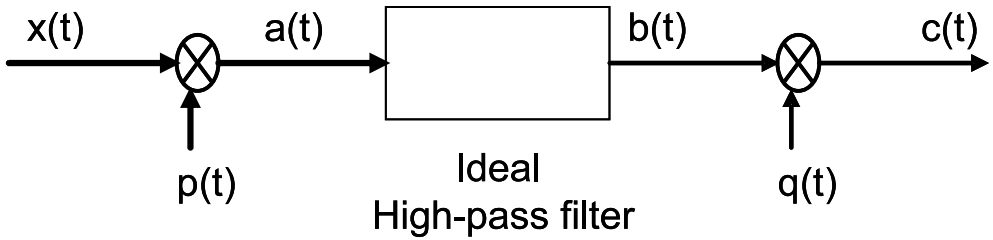
- $Y(f)$ is sketched in the figure below:



- In order to avoid overlap:

$$f_2 - 100 \geq f_1 + 250 \quad \Rightarrow \quad f_2 - f_1 \geq 350 \ \ (\mathrm{Hz})$$

Exercise 22 (a.1)



- Determine the Fourier transform of $x(t) = \frac{\sin(4\pi t)}{\pi t}$:

$$x(t) = \frac{\sin(4\pi t)}{\pi t} = 4\frac{\sin(4\pi t)}{4\pi t} = 4\mathrm{sinc}(4t) \Rightarrow X(f) = \mathrm{rect}\left(\frac{f}{4}\right)$$

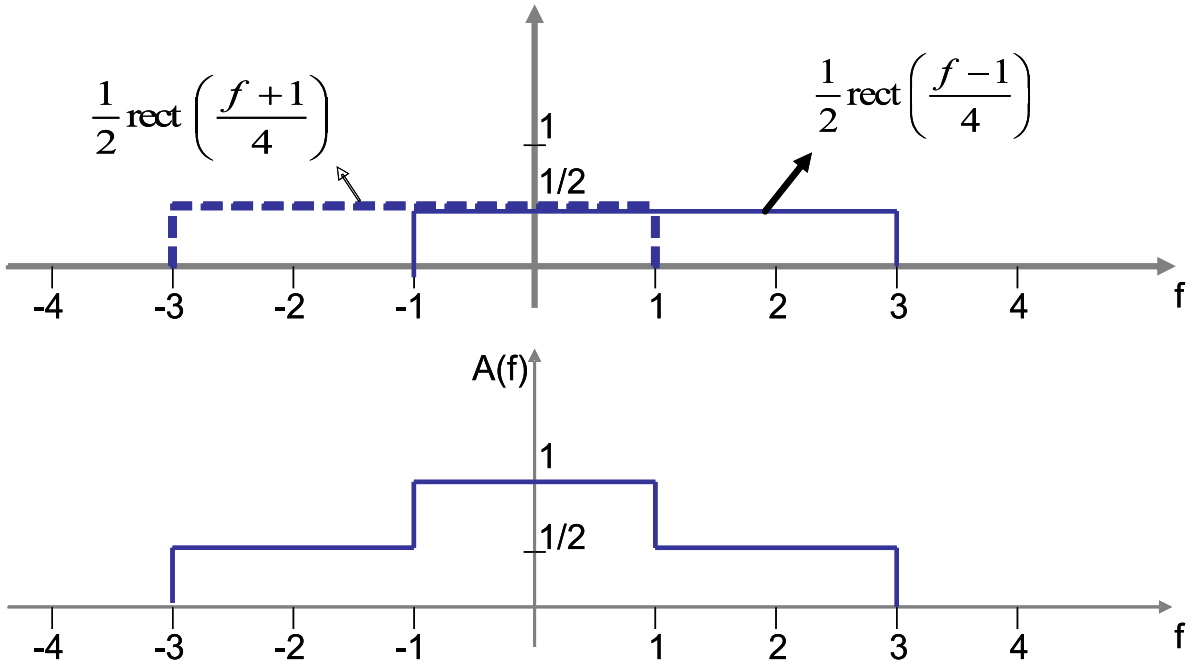
- Determine the Fourier transform of $p(t) = \cos(2\pi t)$:

$$p(t) = \cos(2\pi t) \Rightarrow P(f) = \frac{1}{2}\delta(f-1) + \frac{1}{2}\delta(f+1)$$

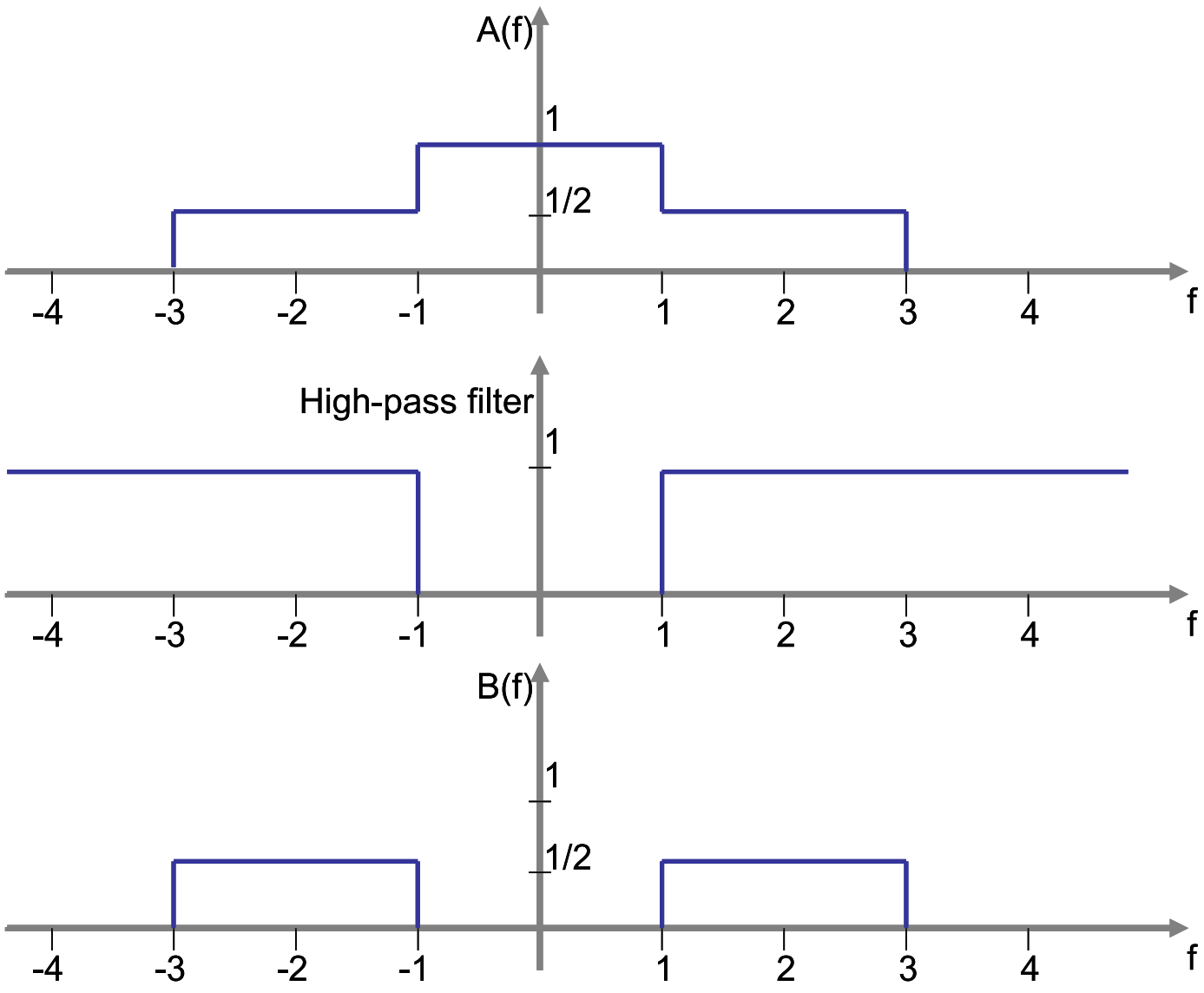
Exercise 22 (a.2)

- Determine the Fourier transform of $a(t)$:

$$
\begin{aligned}
a(t) &= x(t)p(t) \\
A(f) &= X(f) * P(f) \\
&= \frac{1}{2}\mathrm{rect}\left(\frac{f-1}{4}\right) + \frac{1}{2}\mathrm{rect}\left(\frac{f+1}{4}\right)
\end{aligned}
$$



Exercise 22 (b)



Exercise 22 (c)

- $B(f)$ and $b(t)$ can be determined from:

$$B(f) = \frac{1}{2}\text{rect}\left(\frac{f-2}{2}\right) + \frac{1}{2}\text{rect}\left(\frac{f+2}{2}\right)$$

$$\begin{aligned}\Rightarrow \quad b(t) &= \frac{1}{2}2\text{sinc}(2t)e^{j4\pi t} + \frac{1}{2}2\text{sinc}(2t)e^{-j4\pi t} \\ &= \text{sinc}(2t)\left[e^{j4\pi t} + e^{-j4\pi t}\right] \\ &= 2\text{sinc}(2t)\cos(4\pi t)\end{aligned}$$

- On the other hand:

$$q(t) = \frac{\sin(2\pi t)}{\pi t} = 2\frac{\sin(2\pi t)}{2\pi t} = 2\text{sinc}(2t)$$

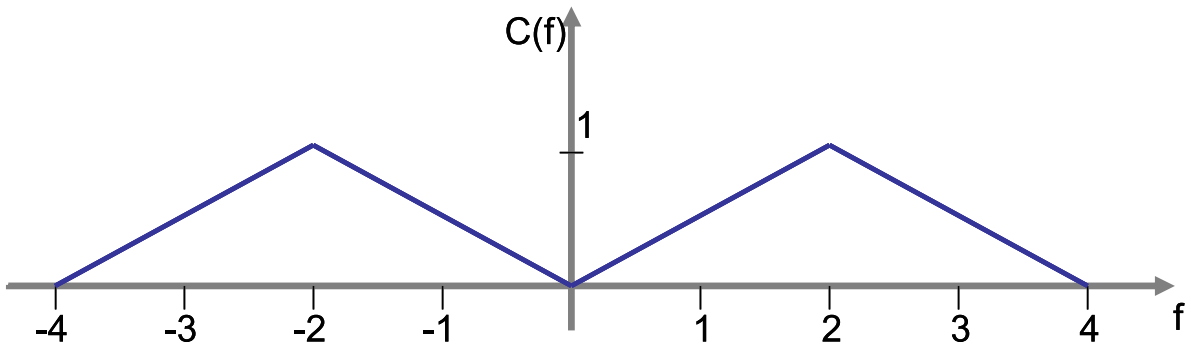
- Consequently:

$$c(t) = q(t)b(t) = 4\text{sinc}^2(2t)\cos(4\pi t)$$

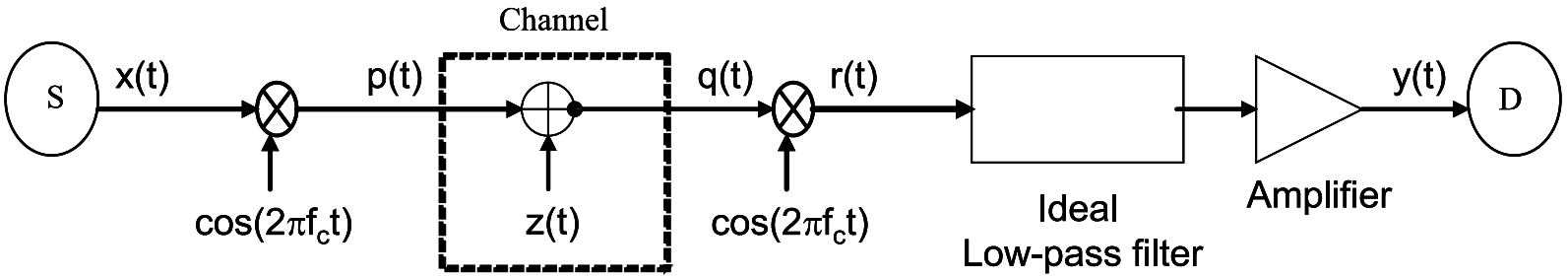
Exercise 22 (d)

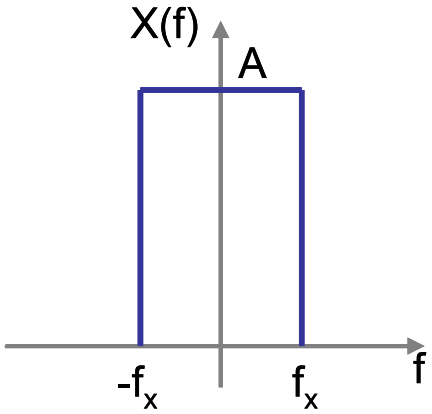
- $C(f)$ can be calculated as follows:

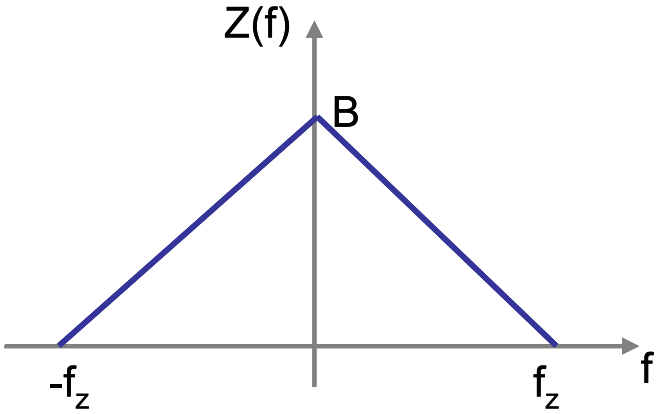
$$
\begin{aligned}
c(t) &= 4\mathrm{sinc}^2(2t)\cos(4\pi t) \\
\Rightarrow \quad C(f) &= 4\frac{1}{2}\mathrm{tri}\left(\frac{f}{2}\right) * \left[\frac{1}{2}\delta(f-2) + \frac{1}{2}\delta(f+2)\right] \\
&= \mathrm{tri}\left(\frac{f-2}{2}\right) + \mathrm{tri}\left(\frac{f+2}{2}\right)
\end{aligned}
$$



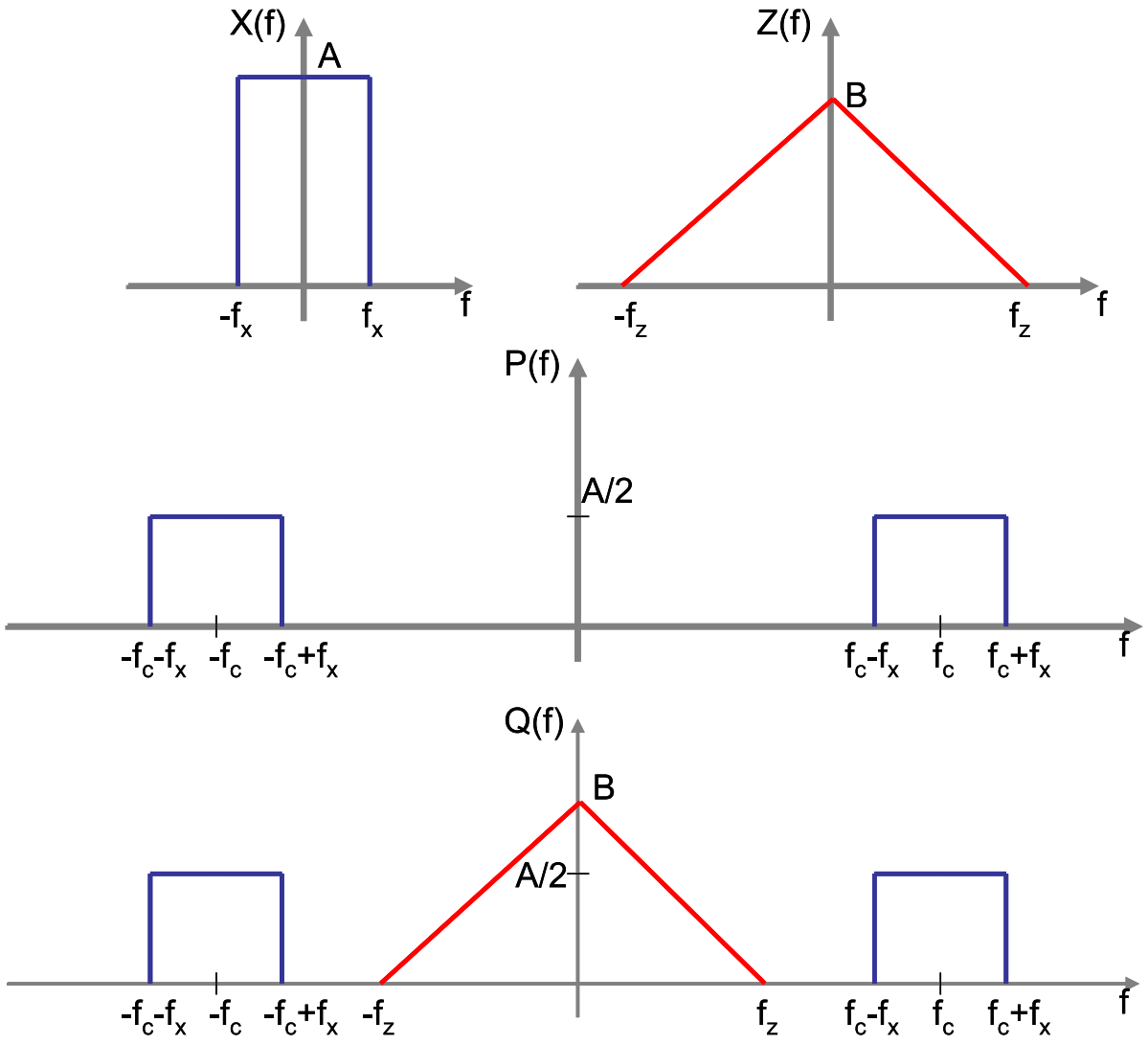
Exercise 23 (1)





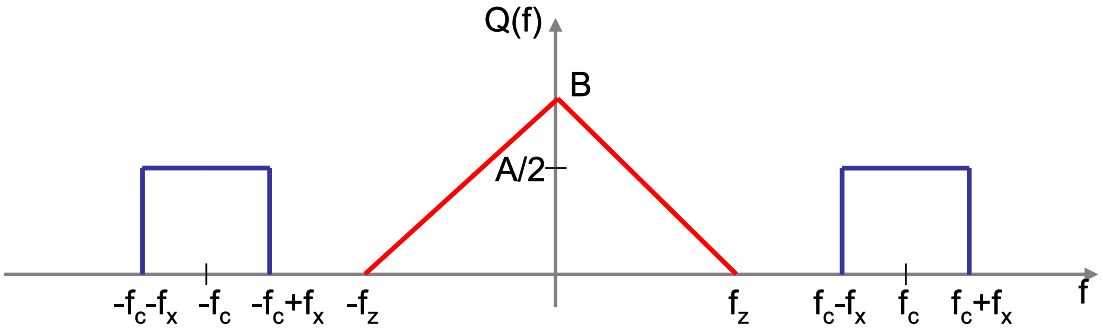


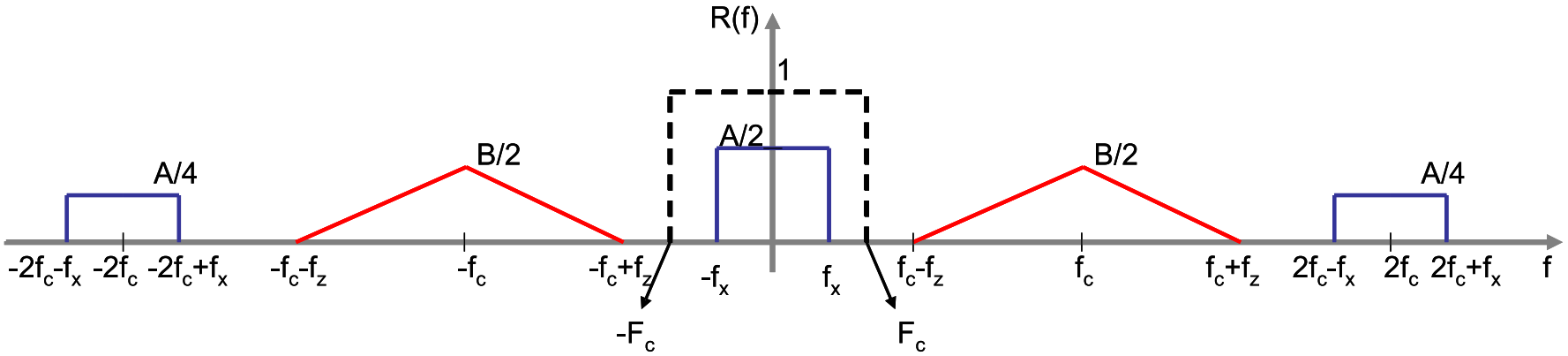
Exercise 23 (2)



- $f_c - f_x \geq f_z \Rightarrow f_c \geq f_x + f_z$

Exercise 23 (3)





- F_c must verify: $f_x \leq F_c \leq f_c - f_z$
- Since the low frequency components have an amplitude of $\frac{A}{2}$ (rather than A), the gain of the amplifier must be equal to 2.